The Hawai‘i Cesspool Prioritization Tool and the Act 217 Sewer Expansion Overlay

University of Hawai‘i Water Resources Research Center

University of Hawai‘i Sea Grant College Program

Presenter:

Christopher K. Shuler,

Contact: cshuler@hawaii.edu

Cesspools harm the people and places we care about



Excess Nutrients Harm Coastal Ecosystems: N causes eutrophication and algal blooms[1]. Linked to decline in coral cover and reduced species diversity[2]. Linked to coral growth anomalies[3].

Effects: Less limu, coral, fish & other marine life. Hurts recreation & cultural practice, threatens economy

How we know: [1]Dailer et al. 2010; [2]Amato et al. 2016; [3]Aguiar et al., 2023 many others



Polluted Waters Put People at Risk: Cesspools linked to increased Fecal Indicator Bacteria[1]. Cesspools release pathogens causing gastroenteritis, Hep-A, salmonellosis, leptospirosis[2].

Effects: Hawai‘i's rate of MRSA infections is double the national average[3]. Swimmers at elevated risk[4].

How we know: [1]Berg et al. 2023; [2]DOH, 2025; [3]Chaiwongkarjohn et al., 2011; [4]Fleisher, 2010



Cesspools Contaminate Drinking Water: Cesspools cause high nitrate (NO3-) levels in drinking water wells[1], and increase Fecal Indicator Bacteria[2]

Effects: NO3- causes Blue Baby Syndrome[3], chronic exposure linked with gastric cancer[4].

How we know: [1]DOH, 2018; [2]Whittier, 2014; [3]Donahoe, 1949; [4]Picetti et al., 2022

Are all cesspools created equal?

Some are worse than others

No One-Size Solution: Treatment effectiveness differs by location, making consistent management difficult

Not a One-Size Problem: Existing regulations acknowledge certain areas face higher environmental risk i.e. coastal areas/wells protected by SMA designations

The Devil is in the Dose: Concentration of contaminants is the critical factor

It's Complicated: Modeling concentrations of nutrients and pathogens is computationally intensive and not practical.

Effective Management requires a simple yet robust framework to assess risk

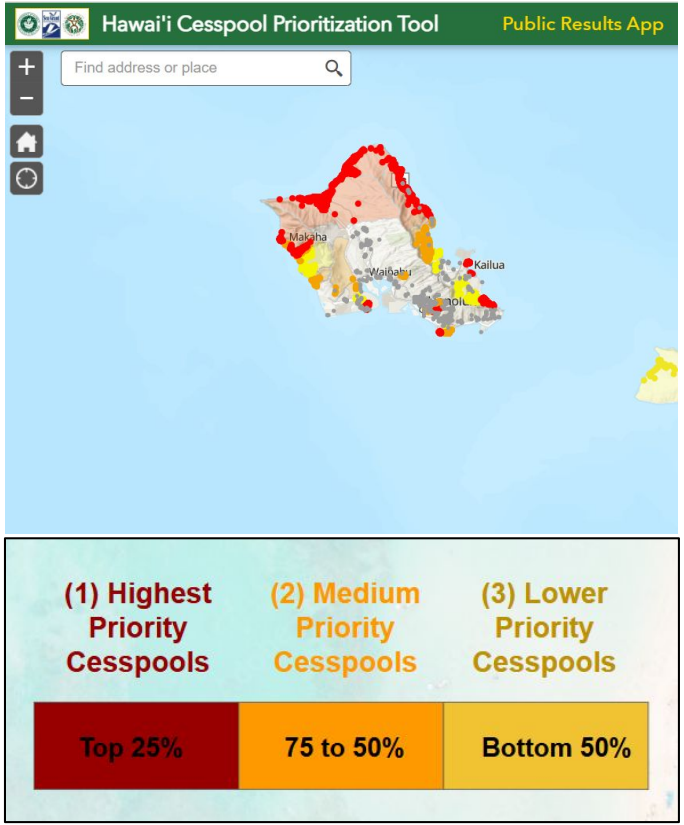


The Hawaii Cesspool Prioritization Tool (HCPT)

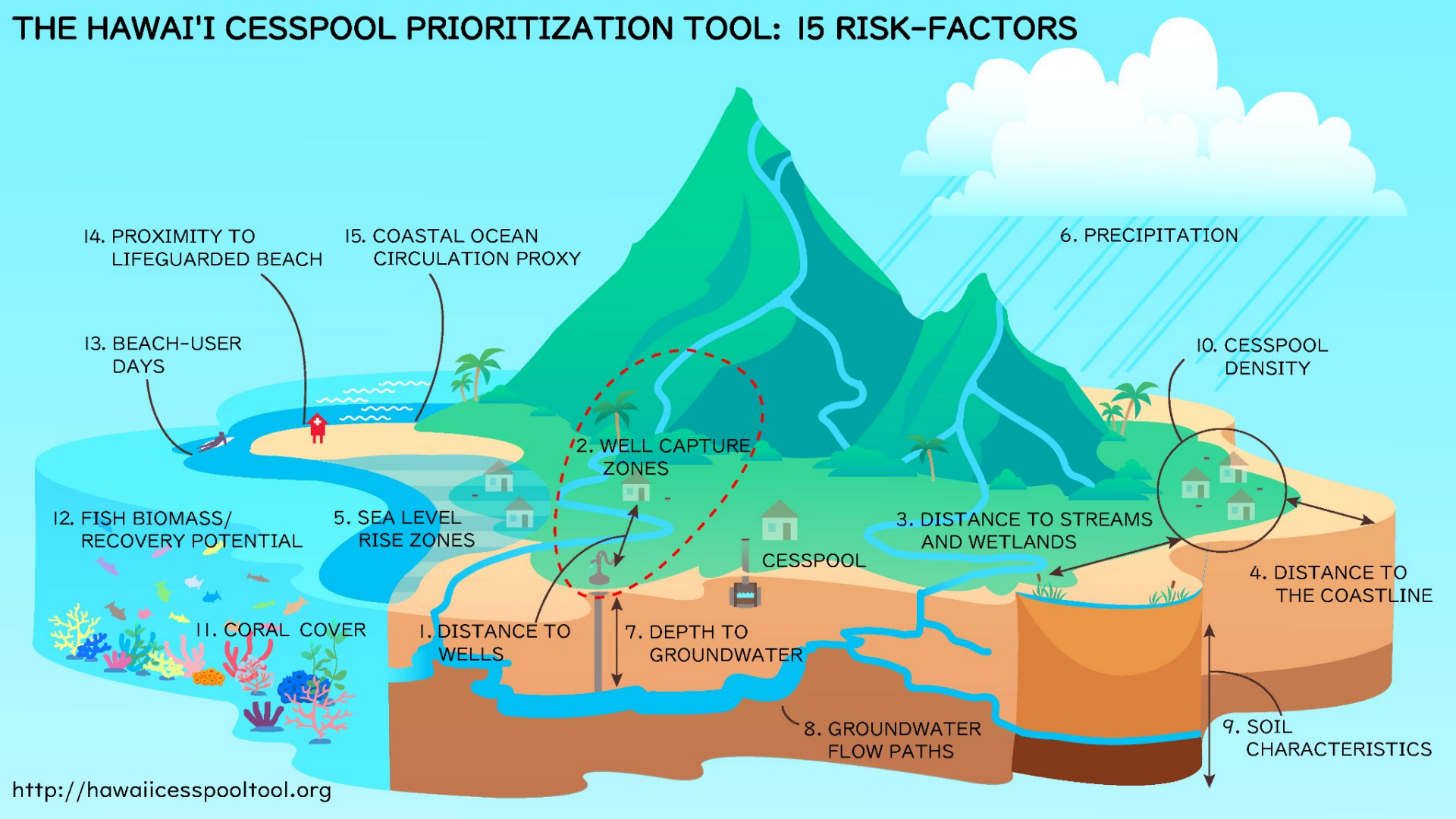
That framework is the HCPT

The HCPT is a decision support tool ranks all of Hawaii's cesspools based on their individual and collective impact to human health and the environment

It considers 15 risk factors and uses a transparent, reproducible, and unbiased method to rank each cesspool in the state into 3 simple categories based on its risk of impacting human and environmental health



THE HAWAI'I CESSPOOL PRIORITIZATION TOOL: 15 RISK-FACTORS



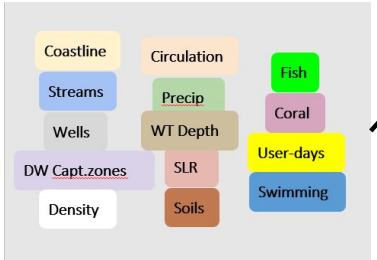
http://hawaiicesspooltool.org

Data Pipeline

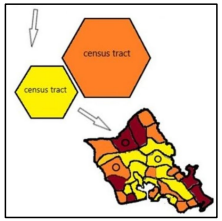
Input Datasets



Cesspool Locations



15 Risk-factor Layers



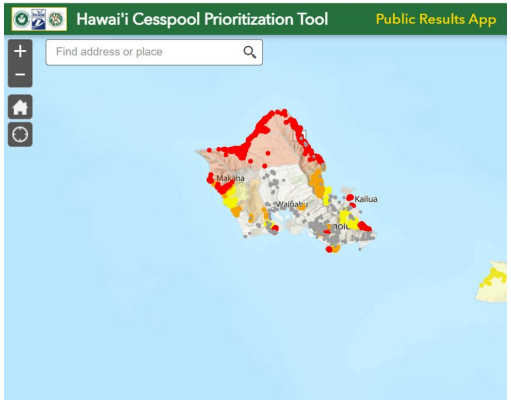
Geographic Boundaries

Workflow

jupyter

Custom geospatial functions

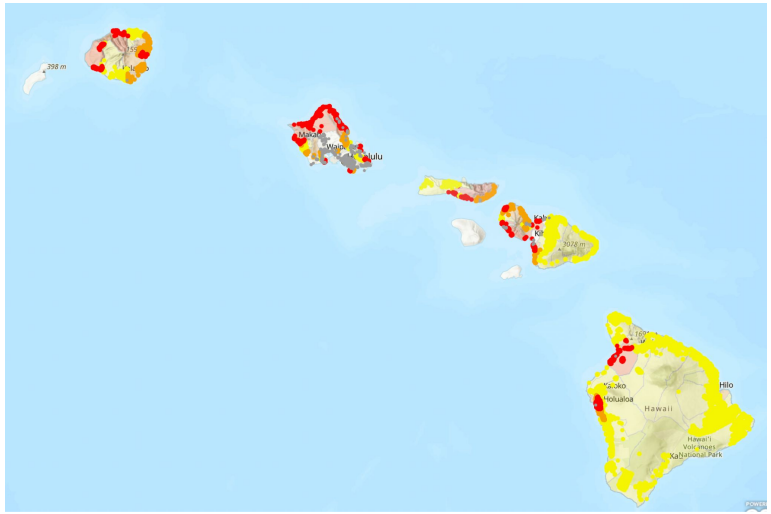
Geographic Outputs and Simple Priority Ranks



(1) Highest Priority Cesspools	(2) Medium Priority Cesspools	(3) Lower Priority Cesspools
Top 25%	75 to 50%	Bottom 50%

Prioritization Results

Transparent (publicly available) and reproducible calculations reduce the complexity of all 15 factors into three data-driven categories



(1) Highest Priority Cesspools	(2) Medium Priority Cesspools	(3) Lower Priority Cesspools
Top 25%	75 to 50%	Bottom 50%
13,821 units	12,367 units	55,237 units

For more information: Visit the HCPT at: http://hawaiicesspooltool.org/

The Centralized Sewer Planning Gap

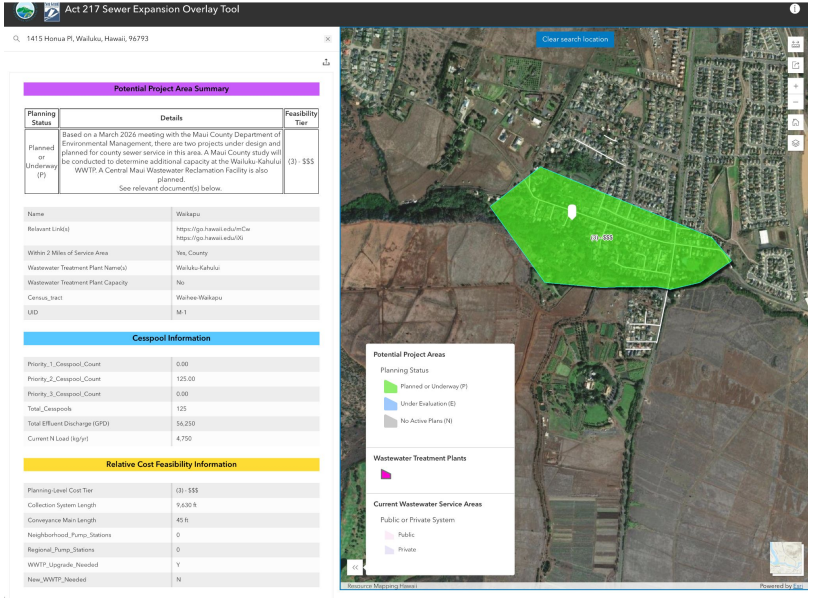


https://www.hawaiicesspooltool.org/

- The HCPT was **not** designed to answer **how** those cesspools should be replaced, by whom, at what cost, or on what timeline.
- No single agency maintains a statewide picture of where sewer expansion is planned, feasible,or likely by 2050, nor has previous legislation addressed centralized sewer planning at the state level.
- Residents remain uncertain whether they should replace cesspools themselves or wait for sewer service. Planners don’t have a comprehensive tool to strategize equitable, state-wide solutions.

Closing The Gap: The Act 217 Sewer Expansion Overlay

- Act 217 of 2024 directed UH WRRC and Sea Grant to develop a centralized sewer expansion overlay for the HCPT.
- The overlay adds to the the existing tool by evaluating where centralized sewer may be feasibly expanded or constructed before 2050.
- Potential expansion areas are evaluated based on their geographic properties, county plans, timing, and cost.

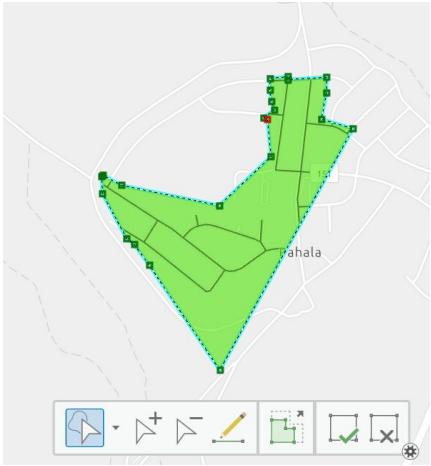


https://seagrant.soest.hawaii.edu/cesspool-tool-wastewater-overlay/

Methods: Data Collection, Engagement, Identifying Project Areas

- Met with county wastewater agencies, DOH, EPA, members of the community.
- Compiled statewide sewer infrastructure and wastewater planning data from county, state, private utility, and PUC: Budgets, plans, engineering studies, and environmental assessments.
- Defined Potential Project Areas: Consistent geographic units for comparing sewer feasibility: Threshold of ≥50 cesspools within 1/8 mi² Limited to conventional gravity sewer expansion, **not** alternative technologies.



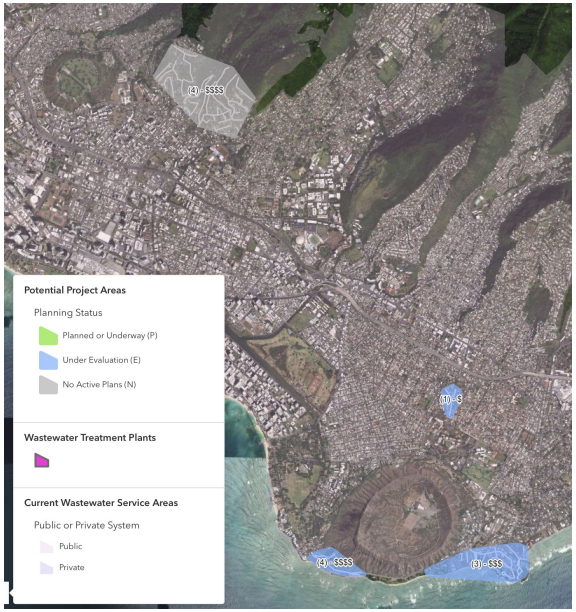


Methods: Planning Status Categories

Planned or Underway (P) Areas where a sewer expansion project has been formally approved, funded, scheduled, or is already under construction or active implementation or confirmed by the responsible wastewater utility provider as expected to be built before 2050.

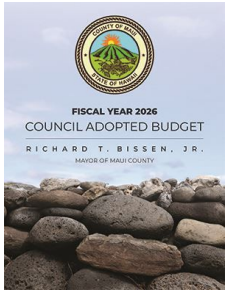
Under Evaluation (E) Areas where centralized sewer expansion has been identified in planning documents, studies, published capital improvement program materials, or county or utility discussions, but has not been formally approved, funded, or scheduled for implementation. These areas remain of uncertain status.

No Active Plans (N) Areas where no known sewer expansion project, study, proposal, or planning effort currently identifies the area for centralized sewer infrastructure. These areas contain cesspools at sufficient density to make centralized wastewater service a potential future option, but no documented planning activity was found at the time of analysis.



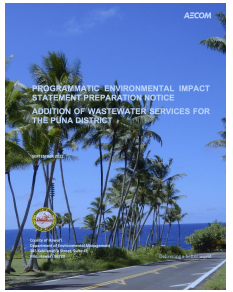
Methods: Cost Modeling

- Developed planning-level relative cost estimates for each Potential Project Area using Hawai‘i-based unit costs and GIS-derived site characteristics.
- Assigned five cost tiers **($ to $$$$$)** based on estimated capital cost per cesspool replaced.
- Validated estimates against currently underway County commissioned engineering reference studies.

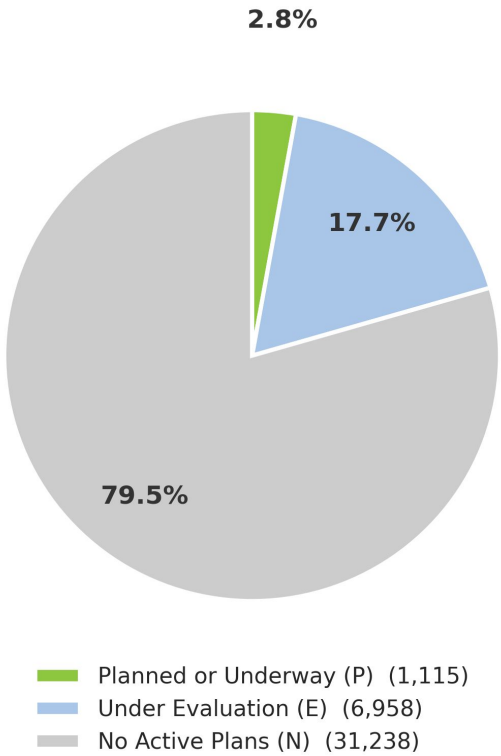






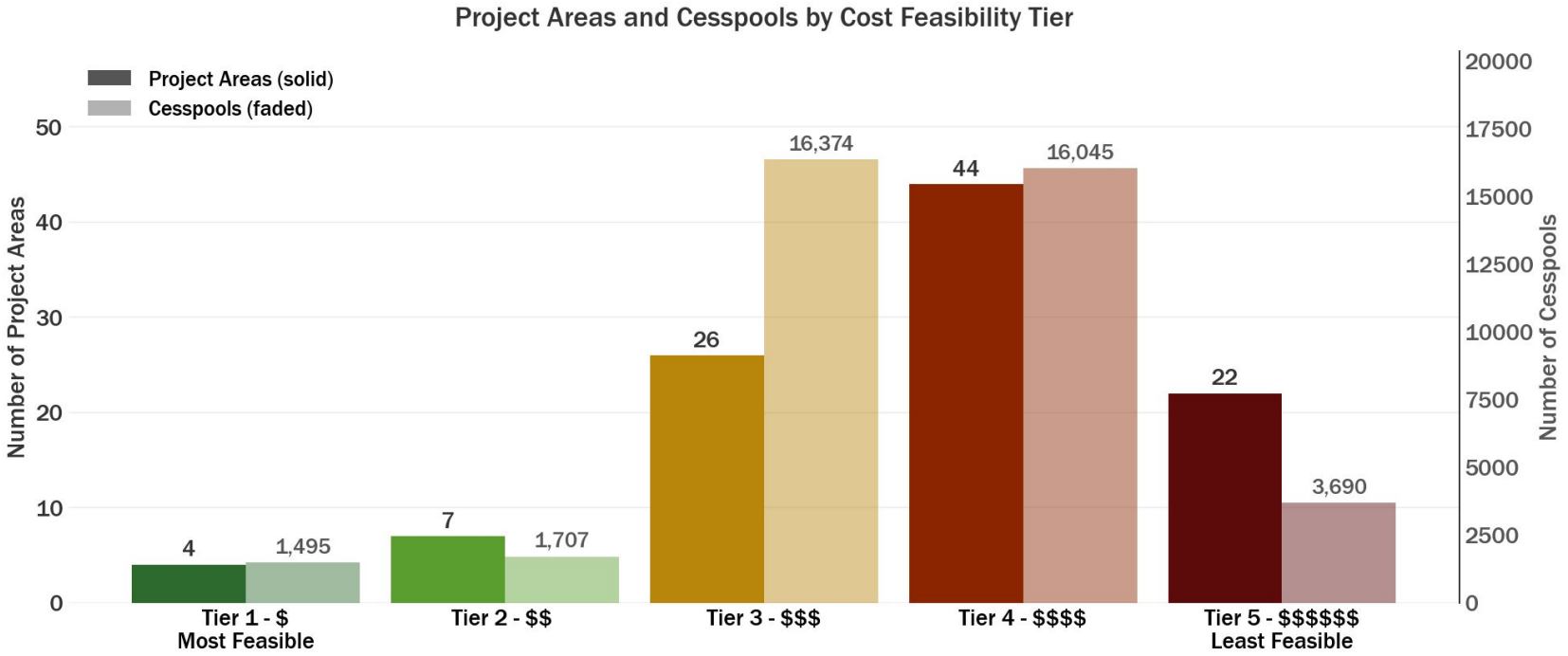


Key Finding 1: Few Active Sewer Projects Planned or Underway



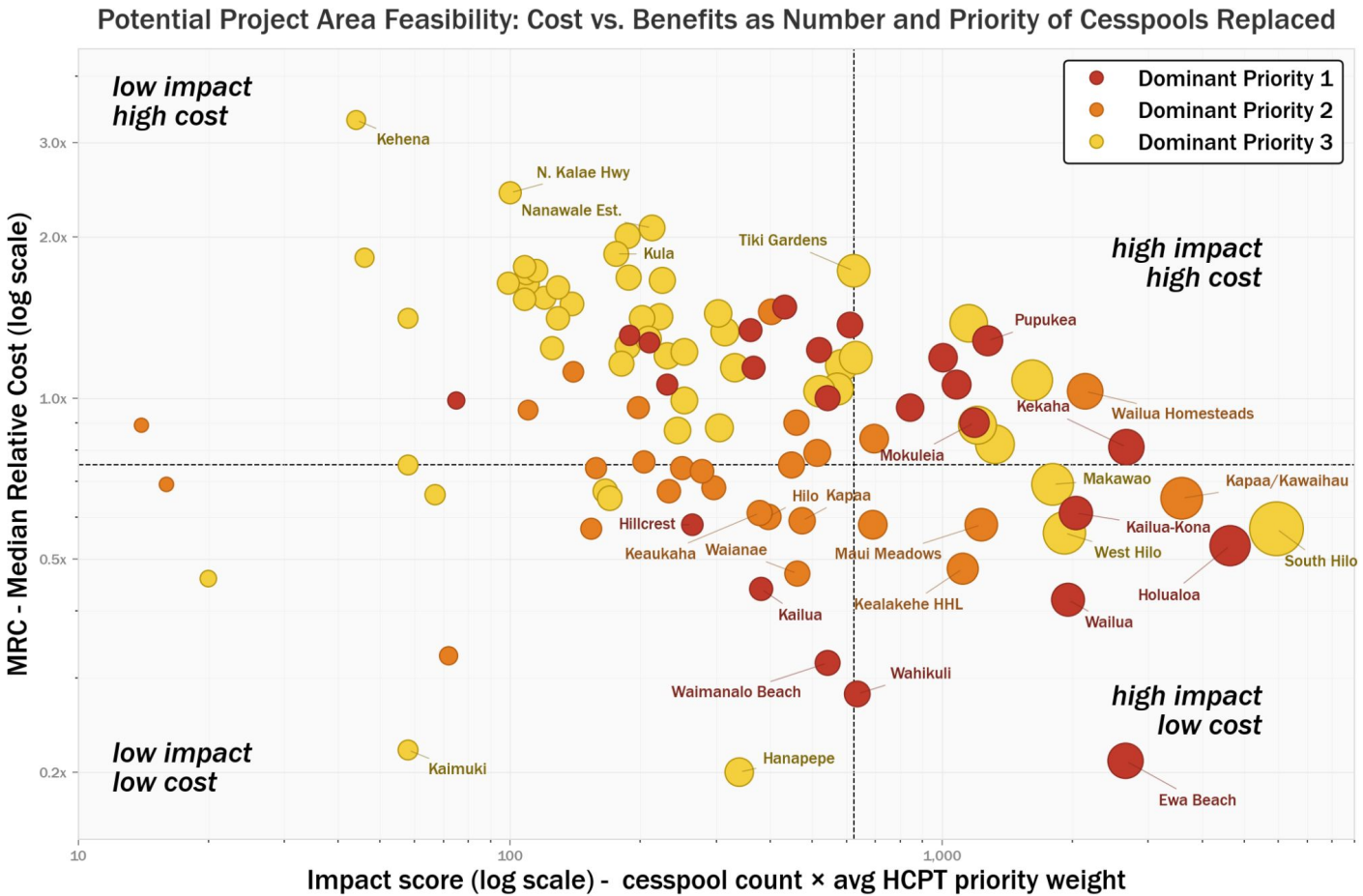
Planning Status	Project Areas	# of Cesspools	% of Project Area Total	% of Statewide Total
Planned or Underway (P)	4	1,115	2.8%	1.3%
Under Evaluation (E)	32	6,958	17.7%	8.3%
No Active Plans (N)	67	31,238	79.5%	37.4%
Total	103	39,311	100.0%	47.1%

Key Finding 2: Feasibility Is The Exception - Most Projects are Expensive



Feasibility Tiers express relative costs, and translate to actual costs as $$ ≈ 3-5x and $$$$ ≈ 8-14x more expensive than a typical on-site alternative

Key Finding 3: Cost and Hazard Data ID Highest Return Opportunities



MRC of 1x ≈ 10x more expensive than a typical on-site alternative

Key Finding 4: Sewer Feasibility Varies Significantly Across Islands

Planning activity does not track need

Scale and feasibility are inversely related

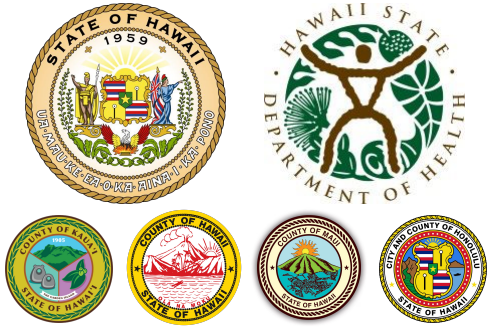
Existing Infrastructure is strongest predictor of feasibility

Island	Total Areas	Number of Potential Project Areas In Cost Tier:					Total Cesspools	Median MRC
		Tier 1	Tier 2	Tier 3	Tier 4	Tier 5		
Hawai'i Island	42	-	1	10	16	15	20,003	1.29
Maui	16	1	1	6	4	4	5,103	0.82
Moloka'i	2	-	-	-	1	1	381	1.69
O'ahu	22	2	4	4	11	1	4,425	0.9
Kaua'i	21	1	1	6	12	1	9,399	0.84
Total	103	4	7	26	44	22	39,311	1

Lessons Learned: Cesspool Conversion Requires Coordinated Action

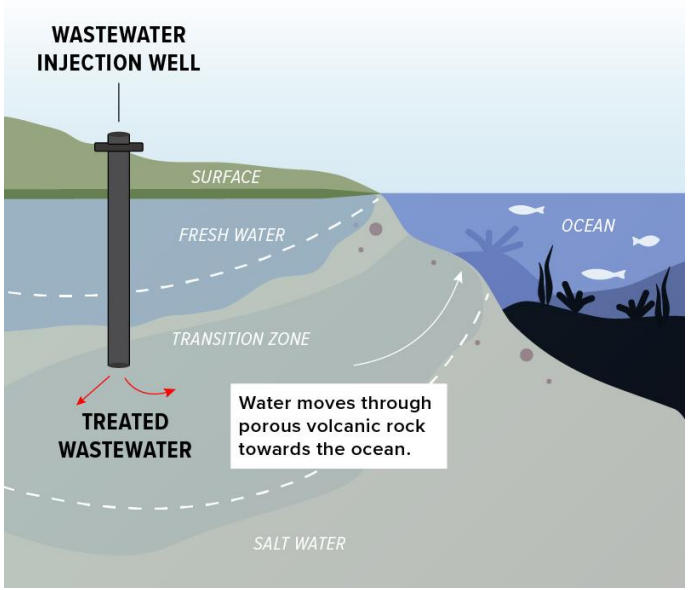
Without coordination among the institutions responsible for regulation, funding, infrastructure planning, and implementation, even the most feasible projects will not advance

- State law establishes the mandate for conversion
- Counties, private companies, or communities carry the responsibility for planning, financing, constructing, operating, and maintaining wastewater systems.
- A dedicated cesspool conversion program within DOH could coordinate across agencies, track conversion progress, and keep statewide data current



Lessons Learned: Disposal of Treated Wastewater Remains an Issue

- County officials indicated new ocean outfalls are extremely difficult to permit, making them unlikely as a practical option for future wastewater expansion.
- Capacity and legal status of injection wells make them another significant limiting factor.
- R-1 (recycled water) regulations do not mandate nutrient reduction
- A sewer project that cannot resolve its disposal pathway cannot move forward regardless of how favorable its collection and conveyance costs appear.



https://civilbeat.org/2024/10/maui-residents-say-its-time-to-stop-letting-treated-wastewater-flow-into-the-ocean/

Lessons Learned: Sewer Expansion Appears Limited to Infill Areas. Existing Funds Focused on Deferred Maintenance and Legally Mandated Projects.



- Most viable near-term candidates are infill areas adjacent to existing infrastructure, where costs are lowest
- Maintenance of aging infrastructure, and fulfilling legal mandates currently absorb available county wastewater funding.
- Centralized sewer is likely not a realistic statewide solution for the majority of cesspools

Other Lessons Learned

Different Centralized Sewer Technologies Can Obscure the Full Costs of System Ownership
Low-pressure systems reduce upfront costs but shift long-term maintenance burdens into the future, and sometimes onto homeowners rather than trained county staff.

Community-Driven Action Has Potential to Advance Cesspool Conversion
Organized communities with federal funding access have driven sewer solutions in Hanalei, Māʻalaea, and Puakō where county planning has not reached.

Private Wastewater Providers Could Play a Role but Face Real Economic Barriers
Private sector participation in cesspool conversion is possible but will require regulatory reform and new funding mechanisms to overcome the economics that currently make it unviable

Climate Change Adds Cost Uncertainty
Sea level rise, intensifying storms, and groundwater changes represent material cost risks to wastewater infrastructure that this model could not capture.

Contact:
Chris Shuler cshuler@hawaii.edu

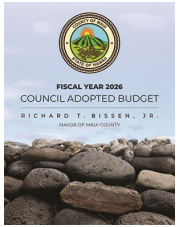
For more information:
Visit the HCPT at:
http://hawaiicesspooltool.org/

Acknowledgements
Aimee Schriber, Autumn McCarter, Johann Lall, Robert Litt

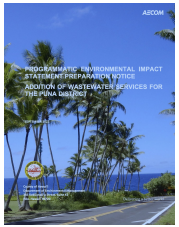
Cost Modelling Appendix: Data Derived Unit-Cost Components

Cost model combines separate unit-costs into major infrastructure components:

- **Collection mains** - Sewer lines used to collect wastewater within neighborhoods
- **Conveyance mains** - Trunk lines used to bring wastewater to treatment facilities
- **Pump stations** - Sized pumps used to overcome gravity or distance
- **Treatment Plants** - Upgrade or new build of WWTPs scaled to facility size



Parameter	Value	Units	Basis
Collection cost per LF	$1,902	$/LF	Derived Value
Conveyance cost per LF	$3,302	$/LF	Derived Value
Treatment, small (<=1,000 connections)	$146,913	$/conn.	Derived Value
Treatment, medium (1,001 to 10,000 connections)	$72,751	$/conn.	Derived Value
Treatment, large (>10,000 connections)	$18,668	$/conn.	Derived Value
Pump, neighborhood	$750,000	$/each	Planning Value
Pump, regional	$6,533,333	$/each	Derived Value



Appendix: GIS-Based Project Area Physical Characteristics

Four physical inputs derived per Potential Project Area using GIS:

- **Collection pipe length -** Road network length as sewer main proxy
- **Conveyance distance -** Shortest network path to nearest treatment facility
- **Pump station counts -** Slope-directed network analysis, calibrated to AECOM Puna plan
- **Treatment capacity status -** County/Private consultation and planning documents

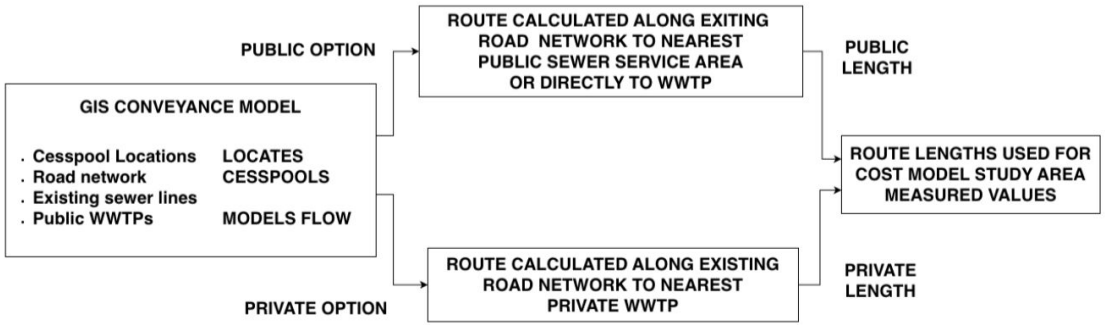


***Figure A1**. Simplified workflow for the GIS-based conveyance routing using GIS network analysis.*

Appendix: Cost Model Outputs

Costs presented as **relative tiers**, not dollar estimates. Underlying model accuracy is at or below AACE Class 5 standards (-50% to +100%), making absolute dollar amounts misleading

- **Five tiers** from (1) - $ (most feasible) to (5) - $$$$$ (least feasible)
- **Tier thresholds** loosely correspond to multiples of typical onsite conversion cost
- **MRC (Median Relative Cost)** calculated for quantitative analysis = area cost divided by statewide median cost.

Table A2. *Relative cost feasibility tiers and MRC range applied to all 103 Potential Project Areas.*

Rating	Cost per Connection	Median Relative Cost (MRC) Range	Approx. Multiple of Onsite Alternative	Areas	Share
(1) - $	Under $120000	Under 0.3	Under 3x	4	4%
(2) - $$	$120,000-$200,000	0.3–0.5	3x-5x	7	7%
(3) - $$$	$200,000-$320,000	0.5–0.8	5x-8x	26	25%
(4) - $$$$	$320,000-$560,000	0.8–1.4	8x-14x	44	43%
(5) - $$$$$	Over $560000	Over 1.4	Over 14x	22	21%

[1] Onsite system cost assumed at $40,000 per connection as a planning-level baseline as reported in Hawai'i (Ishida, 2023; Azambuja, 2024).

Appendix: Cost Model Validation

Carollo Engineers, Inc. (2026) produced estimated costs per connection for similar Potential Project Areas. Made available for validation.

Points above the 1:1 line indicate this study's costs are higher than Carollo's.

Reflects in part Carollo's designs incorporated mix of gravity and low-pressure sewer.

Overall: This study's cost estimates are very comparable to Carollo's estimates.
- $R^2 = 0.54$

